DO NOW — RECALL

Lesson 073 · Trace Tables and Boundary Testing

Name: _____ Class: _____ Date: _____

1. What does CountPasses count?

2. What is a trace table for?

3. What is a boundary condition?

4. Stack Push 3, Push 7, then Pop. What is popped and what remains?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Array values greater than or equal to40.

2. Model answer

Showing changing variables/decisions.

3. Model answer

A limit at which program behaviour changes.

4. Model answer

Popped 7; the stack contains 3.

By the end, you can…

• A trace table records variable values after each meaningful execution step: i, a[i], condition and result.

• A boundary test probes just below, at and above a threshold. Normal, erroneous and exceptional tests reveal different faults.

• Test early success, late success, absent target, and empty arrays. For this function, −1 distinguishes not found from index0.

Vocabulary — know these terms

trace table — State changes in algorithm execution.

sentinel — Special value indicating a condition.

boundary test — Values at or near limit.

exceptional test — Unusual or error-causing case.

NEW LEARNING

Read, annotate and explain

Observe

A trace table records variable values after each meaningful execution step: i, a[i], condition and result.

Explain

A boundary test probes just below, at and above a threshold. Normal, erroneous and exceptional tests reveal different faults.

Evaluate

Test early success, late success, absent target, and empty arrays. For this function, −1 distinguishes not found from index0.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace first even index in [3,5,8,10].

Explain

The function exits at first even occurrence and does not examine10.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace FindFirstEven([7,9,6,4]) with index and parity. [6]

Question 2

Identify expected outputs for [], [2,3], [3,5,8], [1,7,9]. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A program incorrectly returns0 when there is no even value. Explain the ambiguity and propose a safer fix, with tests. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Returning0 is a safe 'not found' value for zero-indexed arrays.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.